Shaders

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1 Introduction

It is assumed that the 3D modeling package used is 3DSMax and the package used for authoring textures is Adobe Photoshop.

A word about the examples in this/these documents.

We used the modeling package 3D Studio max to create Rail Simulator, so this product appears in the examples we have provided. You can, of course, use other packages such as Blender or 3D Canvas. The package we used for authoring textures is Adobe Photoshop, again you may use one of the many other packages available. Regardless of the packages you use, the document should be used as a general guide to the processes to follow to build assets for Rail Simulator

Important note: For your chosen package, you may need additional plug-ins to export the assets into Rail Simulator. We are in the process of finalising these tools and so plug-ins for some packages may not be available immediately.

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2 Dummy Texture

On some shaders, there is a requirement for multiple texture pass effects. An example would be a reflection mapped shader, where the reflection map is applied as a secondary render pass at run-time.

Some of the shader descriptions listed within this document make references to a 'dummy texture' in one or more of the texture slots. These extra texture slots are present to allow the code to apply one of these multi-pass effects at runtime.

It is therefore irrelevant what textures are referenced in these dummy texture slots (as they will get replaced regardless).

However: To ensure that the shaders are kept consistent and simple, it is recommended that a single small texture is created (called for example "dummy.ace") and all references to dummy texture slots use this single texture.

For example:

TrainEnv

Texture slot 1 – diffuse texture
Texture slot 2 – dummy texture

Diffuse texture

Dummy texture

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3 Shaders

There are two types of shader used; Non-FX shaders and FX shaders

3.1 Non-FX shaders

These shaders are less complex than the FX shaders and tend to have fewer texture passes. As a result they can appear more basic in-game.

3.1.1 Unlit object

Tex	
Description	This shader is unlit (it never darkens).
Example use	Interior of coaches, and building windows at night (opaque).

3.1.2 General clutter object

TexDiff	
Description	This shader is lit by diffuse lighting.
Example use	Small clutter objects (that do NOT require shadow maps)

3.1.3 Windows

BlendATexDiff	
Description	This shader uses alpha blending.
Example use	Windows on a train.

3.1.4 Light glows

AddATex	
Description	This shader uses additive alpha.
Example use	Light glows on the ground.

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3.2 FX shaders

These shaders are more complex than the non-FX shaders and tend to have more texture passes. As a result they can appear to have more depth and overall a more interesting/realistic surface in-game.

3.2.1 Stencil Shadows

StencilShadow	
Description	This shader is used specifically for the in-game stencil shadows. For this shader to function correctly, there are strict guidelines which must be adhered to when authoring shadow shapes. Please see authoring guides.
Example use	Stencil shadows

3.2.2 Foliage

TrainFlora TrainViewFacingFlora TrainUprightViewFacingFlora	
Description	These are solely lit by ambient lighting (no diffuse lighting), and can have viewing facing properties.
Example use	Groups of leaves on foliage

3.2.3 Lofts

LoftTexDiff	
Description	These are specific shaders used for generating in-game lofts. These shaders support transparency if required.
Example use	A brick wall
LoftTexDiffTrans	
Description	These are specific shaders used for generating in-game lofts. These shaders support transparency if required.

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Example use	A tree-line

3.2.4 Reflection Map

TrainEnv	
Description	This shader has a second pass reflection map.
Example use	Shiny glass windows on a building (opaque)

3.2.5 Buildings

TrainLightMapWithDiffuse	
Description	This shader allows for a second pass shadow map which allows the asset to have a pre-lit appearance. This helps the give the asset lighting more depth.
Example use	Buildings and Stations

3.2.6 Water

WaterCubeMap	
Description	This shader can be used to generate water surfaces.
Example use	Water lofts

3.2.7 Rolling Stock

TrainBumpSpecEnvMask	
Description	Shiny bumpy metal
Example use	The sides of a locomotive (slightly beaten in appearance)

TrainSpecEnvMask	
Description	Shiny smooth metal
Example use	The sides of a locomotive (smooth in appearance)

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3.2.8 General clutter object (FX shader)

TrainBasicObjectDiffuse	
Description	FX shader version of the basic shader
Example use	Small clutter objects (that do NOT require shadow-maps)

3.2.9 Characters

SkinDiffuse	
Description	Skin shader specifically written for in-game characters. There are guidelines which must be followed when creating characters to ensure the character skeleton matches the expected skeleton within the shader
Example use	Skinned characters or wildlife

3.2.10 Sky

TrainSkyDome	
Description	Skin shader specifically written for creating dynamic skies.
Example use	Skies

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4 Material Properties

4.1 Single sided or double sided?

The Material plug-in supports polygons with single sided or double sided properties. The non-FX and FX shaders handle this property slightly differently.

4.1.1 Non-FX shader

Non-FX shaders render polygons double sided by default.
If using a Non-FX shader and single sided polygons are required, then the 'back-face cull' flag needs to be checked in the material.

4.1.2 FX shader

FX shaders render polygons single sided by default.
If using an FX shader and double-sided polygons are required, then the '2-sided' flag needs to be checked in the material.

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